Satisfaction is independent of the assignment off the free variables

The Open Logic Project contributors

openlogic:first-order-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub prop:valindep; the tagitem/indcase/iftag conditional blocks always render the author's complete branch (preamble tag-macro fix 2026-10-13); upstream typo in the universal case: s2' is defined as Subst{s}{m}{x} where Subst{s_2}{m}{x} is meant (the following sentence confirms 'x-variants of s1 and s2') - author's text delivered verbatim, flagged to the correction layer.

Editorial note. The label prop:satindep names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Value independence of variable assignments for terms. Same unit, \olla-bel{prop:valindep} (source line 432, registered card). Base cases: term values coincide under s1 and s2, driving the atomic and equality cases.

Source reference. Original proposition prop-valindep: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 432–432. Prerequisite pointer invoked at proof line(s) 493; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If the free variables in A are among x_1, …, x_n, and $s_1(x_i) = s_2(x_i)$ for $i = 1$, …, n, then $\mathfrak{M}, s_1 \vDash A$ iff $\mathfrak{M}, s_2 \vDash A$.*

Proof. We use induction on the complexity of A. For the base case, where A is atomic, A can be: $\top$,$\bot$,$R(t_1, \ldots, t_k)$ for a k-place predicate R and terms t_1, …, t_k, or $t_1 = t_2$ for terms t_1 and t_2. In the latter two cases, we only demonstrate the forward direction of the biconditional, since the proof of the reverse is symmetrical.

1. $A \equiv \top$: both $\mathfrak{M}, s_1 \vDash A$ and $\mathfrak{M}, s_2 \vDash A$.

2. $A \equiv \bot$: both $\mathfrak{M}, s_1 \nvDash A$ and $\mathfrak{M}, s_2 \nvDash A$.

3. $A \equiv R(t_1, \ldots, t_k)$: let $\mathfrak{M}, s_1 \vDash A$. Then

$$\langle \mathrm{val}(t_1)M[s_1], \ldots, \mathrm{val}(t_k)M[s_1] \rangle \in sRM.$$

   For $i = 1$, ..., k, $\mathrm{val}(t_i)M[s_1] = \mathrm{val}(t_i)M[s_2]$ by prop-valindep. So we also have $\langle \mathrm{val}(t_i)M[s_2], \ldots, \mathrm{val}(t_k)M[s_2] \rangle \in sRM$, and hence $\mathfrak{M}, s_2 \vDash A$.

4. $A \equiv t_1 = t_2$: suppose $\mathfrak{M}, s_1 \vDash A$. Then $\mathrm{val}(t_1)M[s_1] = \mathrm{val}(t_2)M[s_1]$. So,

$$\begin{aligned} \mathrm{val}(t_1)M[s_2] &= \mathrm{val}(t_1)M[s_1] && \text{(by prop-valindep)} \\ &= \mathrm{val}(t_2)M[s_1] && \text{(since } \mathfrak{M}, s_1 \vDash t_1 = t_2\text{)} \\ &= \mathrm{val}(t_2)M[s_2] && \text{(by prop-valindep),} \end{aligned}$$

   so $\mathfrak{M}, s_2 \vDash t_1 = t_2$.

Now assume $\mathfrak{M}, s_1 \vDash B$ iff $\mathfrak{M}, s_2 \vDash B$ for all formulas B less complex than A. The induction step proceeds by cases determined by the main operator of A. In each case, we only demonstrate the forward direction of the biconditional; the proof of the reverse direction is symmetrical. In all cases except those for the quantifiers, we apply the induction hypothesis to sub-formulas B of A. The free variables of B are among those of A. Thus, if s_1 and s_2 agree on the free variables of A, they also agree on those of B, and the induction hypothesis applies to B.

1. $A \equiv \neg B$: exercise.

2. $A \equiv B \wedge C$: exercise.

3. $A \equiv B \vee C$: exercise.

4. $A \equiv B \to C$: exercise.

5. $A \equiv B \leftrightarrow C$: exercise.

6. $A \equiv \exists x\, B$: exercise.

7. $A \equiv \forall x\, B$: exercise.

By induction, we get that $\mathfrak{M}, s_1 \vDash A$ iff $\mathfrak{M}, s_2 \vDash A$ whenever the free variables in A are among x_1, ..., x_n and $s_1(x_i) = s_2(x_i)$ for $i = 1$, ..., n. □